

分光计测量物质折射率

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1 测量三棱镜的折射率

φ'_1	φ'_2	φ''_1	φ''_2	δ_{min}^1
248°37'	209°56'	68°37'	29°56'	38°41'
φ'_1	φ'_2	φ''_1	φ''_2	δ_{min}^2
198°54'	160°10'	18°54'	340°10'	38°44'
φ'_1	φ'_2	φ''_1	φ''_2	δ_{min}^3
216°33'	177°48'	36°33'	357°48'	38°45'

$$\delta_{min} = \frac{(\varphi'_2 - \varphi'_1) + (\varphi''_2 - \varphi''_1)}{2}$$

$$\bar{\delta}_{min} = \frac{\delta_{min}^1 + \delta_{min}^2 + \delta_{min}^3}{3} \approx 38^\circ 43'$$

$$n_{\text{三棱镜}} = \frac{\sin \frac{\bar{\delta}_{min} + A}{2}}{\sin \frac{A}{2}} = 1.5176588530685438$$

由于我们三次测量相互独立（即我们测量一次就转了一次刻度盘），我们没办法使用间接测量的误差传递公式进行计算。

我们只能对三次计算出的 δ_{min} 进行 A 类不确定度分析。

$$u_A(\delta_{min}) = \sqrt{\frac{\sum_i^N (\delta_{min}^i - \bar{\delta}_{min})^2}{N(N-1)}} = 0.020030840419245546^\circ$$

$$u_B(\delta_{min}) = \frac{\Delta}{\sqrt{3}} = \frac{\frac{1}{60}}{\sqrt{3}} = 0.009622504486493762^\circ$$

$$u_{\delta_{min}} = np.sqrt(u_A^2(\delta_{min}) + u_B^2(\delta_{min})) = 0.02222222222223267^\circ$$

$$U = \frac{\partial n}{\partial \delta_{min}} u_{\delta_{min}} = \frac{1}{\sin \frac{A}{2}} \cos \frac{\delta_{min} + A}{2} \times \frac{1}{2} u_{\delta_{min}} = 0.000387850944887647$$

则

$$n = 1.5177 \pm 0.0004$$

2 测量水的折射率

分界线 ₁	十字叉丝 ₁	分界线 ₂	十字叉丝 ₂	r'
154°32'	156°47'	334°32'	336°47'	2°15'
分界线 ₁	十字叉丝 ₁	分界线 ₂	十字叉丝 ₂	r'
175°47'	178°2'	355°47'	358°2'	2°15'

分界线 ₁	十字叉丝 ₁	分界线 ₂	十字叉丝 ₂	r'
$197^{\circ}47'$	$200^{\circ}2'$	$17^{\circ}47'$	$20^{\circ}2'$	$2^{\circ}15'$

$$\bar{r}' = 2^{\circ}15'$$

$$\bar{n}_x = \sin A \sqrt{n^2 - \sin^2 r'} + \cos A \sin r' = 1.3335211879605458$$

对于 r' , 由于三次测量的结果相同, 则 A 类不确定度不存在.

$$u_{r'} = u_B(r') = \frac{\Delta}{\sqrt{3}} = \frac{\frac{1}{60}^{\circ}}{\sqrt{3}} = 0.009622504486493762^{\circ}$$

$$U = \sqrt{\left(\frac{\partial n_x}{\partial \delta_{min}}\right)^2 \times u^2(\delta_{min}) + \left(\frac{\partial n_x}{\partial r'}\right)^2 \times u^2(r')}$$

$$\frac{\partial n_x}{\partial \delta_{min}} u_{\delta_{min}} = \sin A \frac{n}{\sqrt{n^2 - \sin^2 r'}} \frac{1}{\sin \frac{A}{2}} \cos \frac{\delta_{min} + A}{2} \times \frac{1}{2} u_{\delta_{min}} = 0.0002188340339156575$$

$$\frac{\partial n_x}{\partial r'} u_{r'} = \left[\sin A \frac{-\sin r' \cos r'}{\sqrt{n^2 - \sin^2 r'}} + \cos A \cos r' \right] u_{r'} = 7.819983676468333 \times 10^{-5}$$

$$u_{n_x} = \sqrt{0.01253826655718989^2 + 0.004480520605228325^2} = 0.00023238663659905717$$

$$n_x = 1.3335 \pm 0.0002$$